W4A16 quantization

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Approach

- **Quantization**

  Symmetric groupwise int4 RTN

  AWQ preprocessing

- **Matmul fused with dequantization**

  Triton W4A16 GEMM for prefill (M > 1)

  CUDA W4A16 GEMV for decode (M = 1)

- **Evaluation on Qwen3-1.7B**

  WikiText-2 perplexity: BF16 / RTN / AWQ

  Per-shape kernel microbenchmarks

  End-to-end prefill + decode roofline vs BF16 and GemLite

References

[1] Frantar, E., Castro, R. L., Chen, J., Hoefler, T., & Alistarh, D. (2025).
MARLIN: Mixed-Precision Auto-Regressive Parallel Inference on
Large Language Models.
Proceedings of the 30th ACM SIGPLAN Annual Symposium on Principles and Practice of Parallel Programming (PPoPP '25).
arXiv:2408.11743

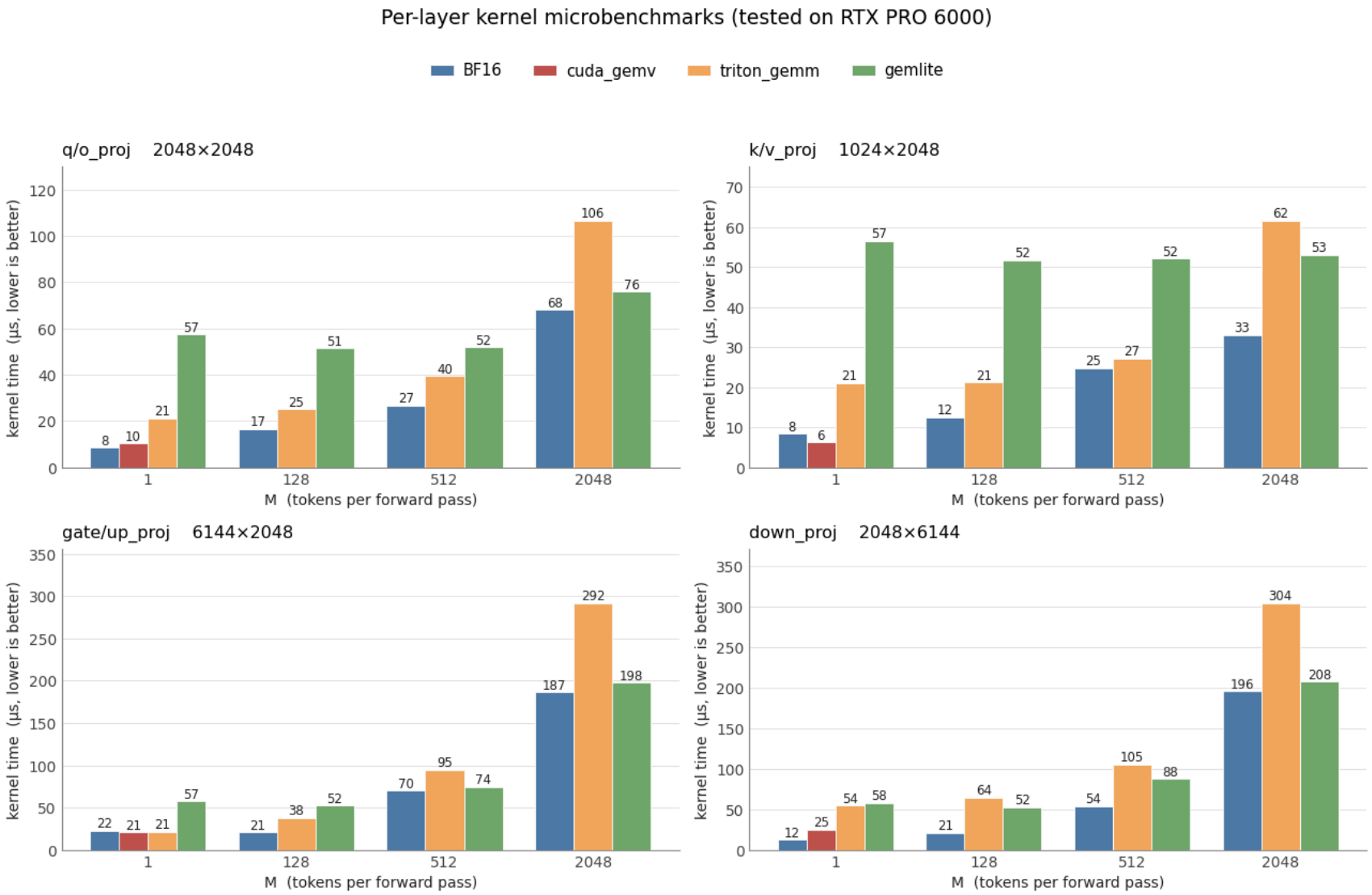
[2] GemLite: Triton Kernels for Efficient Low-Bit Matrix Multiplication.
https://github.com/dropbox/gemlite

Results

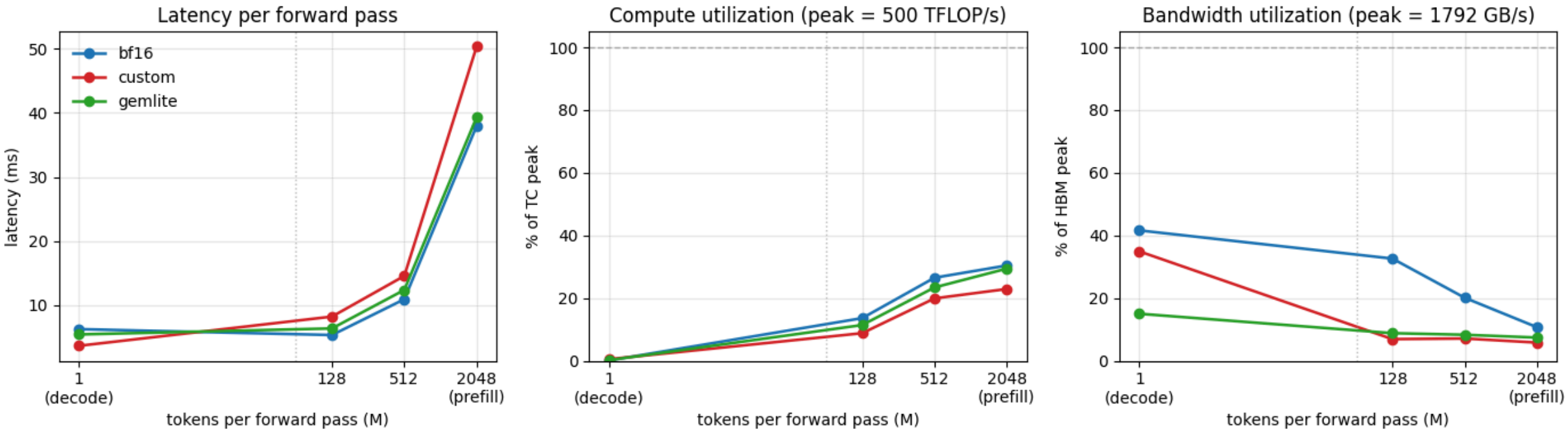
Model	WikiText-2 Perplexity
BF16	16.72
INT4 RTN	22.69
INT4 AWQ *	17.76

* AWQ is applied to all quantized decoder linears except o_proj.

Results


Per-layer kernel microbenchmarks (tested on RTX PRO 6000)
BF16
cuda_gemv
triton_gemm
gemlite
q/o_proj 2048×2048
k/v_proj 1024×2048
gate/up_proj 6144×2048
down_proj 2048×6144
kernel time (µs, lower is better)
M (tokens per forward pass)
1
128
512
2048
8 10 21 57
17 25 51
27 40 52
68 106 76
8 6 21 57
12 21 52
25 27 52
33 62 53
22 21 21 57
21 38 52
70 95 74
187 292 198
12 25 54 58
21 64 52
54 105 88
196 304 208

Results


Latency per forward pass
bf16
custom
gemlite
latency (ms)
50
40
30
20
10
1
(decode)
128
512
2048
(prefill)
tokens per forward pass (M)
Compute utilization (peak = 500 TFLOP/s)
% of TC peak
100
80
60
40
20
0
1
(decode)
128
512
2048
(prefill)
tokens per forward pass (M)
Bandwidth utilization (peak = 1792 GB/s)
% of HBM peak
100
80
60
40
20
0
1
(decode)
128
512
2048
(prefill)
tokens per forward pass (M)

Results

kernel	tpot (ms)	tok/s	% HBM peak
BF16	6.23	161	41.6
GemLite	5.40	186	15.0
Custom	3.63	271	34.9

Per-token latency, throughput, and HBM utilization at decode M = 1.

Conclusions

- **1.7× decode speedup with custom GEMV over BF16**

  271 tok/s vs 161 BF16 / 186 GemLite at M=1

- **AWQ recovers near-BF16 quality**

  WikiText-2 PPL: 16.72 BF16 → 22.69 RTN → 17.76 AWQ

- **W4A16 wins where bandwidth is the bottleneck**

  Decode (M=1): bandwidth-bound → quantization pays off

  Prefill (M ≫ 1): compute-bound → no win without a CUDA tensor-core GEMM